

Q1. Material Grade for:

- a. 8ft W-Beams
- b. 16ft W-Beams
- c. Flared Ends
- d. "T" Steel Beams

Available materials: a. S235JR, b. S275JR & c. S355JR

A1 *MS: S355JR for T beam. Guardrails are required to be 12 gauge steel W beams, as stated in the SOW.

Q2. Confirm for Posts you require "T" sections instead of "H" sections and provide:

- a. Web Thickness
- b. Flange Thickness

A2 *MS: 4x4x7.5 T beams, as indicated in the reference drawing.

Q3. For Guardrails provided by the Contractor, it is stated that they should be provided without holes; and later on drilled. Given the fact that Guardrails will be Hot-dip galvanized, we would appreciate if you could provide the bolt pattern or design standard required for these guardrails. This will allow us to pre-drill the holes (normally oval shape) prior to installation, and without affecting the protective coating.

A3 *MS: Guardrail bolt hole locations are not standardized and vary to suit conditions of the ship. The arrangement details for the guardrails to be replaced are available in the reference drawing, as well as pictures of the current guard rails onboard (see attached). However, when originally installed they were cut and drilled onboard during installation.

Q4. Guardrail accessories, e.g. splice washers, splice bolts/nuts & post bolts, are not mentioned in your specification. Kindly please advise if MMS shall include the above as per common practice (i.e. 1x post bolt + 1x beam washer per post/connection & 8x splice bolts/nuts + 8x splice washers per post/connection)

A4 *MS: Details of the hardware needed are in the reference document, page 3. Splice nut/bolts are 5/8". Post nut/bolt/washers are 5/8". As stated in the SOW all hardware is to be CRES.

Q5. Section 7.5 of the Spec refers to blasting & coating of guardrails/frames/brackets, although Section 3.2 specifies material with hot-dip galvanized surfaces kindly please advise whether HDG is required on the requested material, as we can also supply them without HDG."

A5*MS: The guardrails should be painted with 2 coats of primer followed by one topcoat of safety yellow prior to installation

Please see the attached "USNS Arctic Guard Rails Exterior" file